Limex Quantum Project: Probabilistic Modelling of the Limit Order Book

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Contents

1 Introduction

This chapter motivates the research and findings of this report. We begin by defining and explaining the mechanics of key market mechanisms essential for the operation of modern **capital markets**. Understanding how buyers and sellers interact is essential for explaining how we generate predictive price signals from the most granular of market data.

1.1 Understanding the Limit-Order-Book (LOB)

Level Cumulative Volume (#)	Asset Price ($)	Level Volume (Shares) (#)
$\sum_{i=1}^{n} V_{Ask,i}$	Ask_n	$\sum_{v \in V_{Ask_n}} v$
.	.	.
.	.	.
.	.	.
600	202	500
100	201	100
	fair value	
100	199	100
300	198	200
.	.	.
.	.	.
.	.	.
$\sum_{i=1}^{m} V_{Bid,i}$	Bid_m	$\sum_{v \in V_{Bid_m}} v$

Table 1: A simplified view of a Limit-Order-Book (LOB) with depth (left most column)

The **LOB** is a snapshot of resting unfilled (limit) orders at any point in time, giving us a view of market participation on both the demand side (Bids) and supply side (Asks). Table 1 is a simplified view of a general LOB you will see offered by most charting software or cryptocurrency exchanges. At the center, we have the **fair value price**, which is the valuation of the asset, and is estimated by the average of the top of the book (best bid and best ask). **Price-takers** are aggressive market participants that want instantaneous order execution; these orders will always take **liquidity** (Order volume, Left column Table 1) away from book as they get filled. A price-taker who is buying will, buy the asset at the lowest available ask price. For example a price taker who wants to by 10 shares of stock XYZ at market will pay a price of \$201, taking $10 * \$201$ of volume of the book Table 1 leaving the resting top of the book supply at 90 shares of volume of stock XYZ at \$201. Similarly the same price-taker wanting to exit their current **long position** will do so at the best available bid price, here \$199, taking the same liquidity off the book but now from the demand side. Notice also how we used the terms demand and supply interchangeably to describe Bid and Asks, This is key as it helps us think of the LOB as a market where standard economic theory applies.

1.1.1 Limit Orders

A limit order is an order that guarantees a market participant a better price than the market price. Traders who set limit orders are referred to as **price-makers**. Using Table 1 as an example, all orders above the the \$201 Ask price are sell-side limits, meaning all resting orders at this level are **short orders**, and will be filled when the bid price is greater than or equal to this level. Similarly, all prices below the \$199 bid price are buy-side limits; these are **long orders** that will be filled when the ask price is less than or equal to the level. In both cases, a price better than or equal to the limit price is guaranteed.

1.1.2 Liquidity

This term refers to the quantity of **volume** (value of order) in a market, i.e. the sum of buy and sell orders within a timeframe. As you may have already noticed in our prior examples, price-takers demand liquidity, and price makers provide liquidity; this means it is price-takers that move prices.

1.1.3 Bid-Ask Spread

This is the difference between the lowest price a market participant is willing to sell at and the highest price a market participant is willing to buy at. Subtracting the bid price from the ask price yields the spread. The more liquid a market, i.e., the more volume of orders in the LOB, the lower the spread and the more efficient the market.

1.1.4 Matching-engines and Order Execution Priority

Price	Size	Time (nanoseconds since UNIX epoch)
	.	.
	.	.
	.	.
	10	2025-07-22 08:00:00.000634385+00:00
201	1	2025-07-22 07:05:03.032842821+00:00

Table 2: A visualisation of the time-price priority order matching mechanism using the top of the book ask from Table 1.

We are yet to talk about how orders are sorted within a level, as this will determine the logic in which orders are executed. Electronic markets employ a **matching-engine** to solve the problem of when and what orders are filled. In the standard setting, the exchange will use a **time-price priority** matching engine [1]. This is a **FIFO** (first-in-first-out) based matching engine using price and time as the only criteria to determine when a trade will be facilitated. The first order at any price level, irrespective of its size will be the first order matched. Modifications to an existing order will however, change priority, specifically

- **Increase in size.** The order will be removed and appended to the end of the queue.
- **Change in price.** The order will be deleted from the level and appended to an existing level; otherwise, a new level will be created.
- **Change in account number**. Identical action to increase in size.

Table 2 highlights the time-price priority nature of the LOB at the top of the book ask from Table 1. Time-price priority matching engines ensure orders submitted with the least lag will be executed at the best bid and ask prices. This can lead to High-Frequency-Traders (HFTs) offering a large quantity of unbalanced orders, resulting in large price swings [2].

Many other matching engines exist e.g. **FIFO with LMM**, **Pro-Rate** and **FIFO with Top Order and LMM**, which theoretically claim to result in more "fair markets" [1].

1.1.5 Market-Makers and LOB Nuances

Market makers (MMs) play a special role in markets, as they facilitate matching demand with supply. Market makers make money from facilitating trades

- **Incoming Buy Order.** The MM sells at the top of the book ask, holding a short position and making a profit if the price dips below the ask, incurring a loss otherwise

- **Incoming Sell Order.** The MM buys at the top of the book bid, holding a long position and making a profit if the price rises above the bid, incurring a loss otherwise.

You may be asking, what happens in the case when a buy/sell order is submitted with a price greater/less than the ask/bid. In this case, the order will not rest in the LOB, and will be executed instantaneously at the top of the book, ensuring the LOB is never a crossed book.

2 Task

We are provided with **Level three (L3) market data** for two instruments across several days. Using data aggregation techniques and understanding of the LOB, we have to **reconstruct the LOB** and derive a predictive price signal which we must use to from a trading strategy.

3 Probolistic Model

In this section, we motivate and derive a probabilistic approach to modeling the mechanics of the LOB, which are used to construct a predictive signal. Our signal is focused on modelling order size and order frequency.

3.1 Trade Size

Each incoming trade ,$N_i(t)$, no matter its type (limit or market) has an associated trade size, Q. We make the assumption that larger trades are less frequent to hit the book, specifically we assume the likelihood of trade size is proportional to the trade size raised to some exponent, or more simply the change in trade size likelihood varies as some exponent of trade size. More formally we have

$$\begin{aligned}\mathbb{P}(Q > v) &\propto v^{-k} \\ &= cv^{-k} \text{ for } v \geq v_m\end{aligned} \tag{1}$$

This is a tail probability or survival function. α is some parameter we will estimate via Maximum-Likelihood-Estimation. Noting the fundamental theorem of calculus and the definition of a CDF Function, we can recover the unknown and c.

From the survival function we know

$$\begin{aligned}&\mathbb{P}(Q > v) = 1 - \mathbb{P}(Q < v) \\ &\mathbb{P}(Q < v) = 1 - \mathbb{P}(Q > v) \iff F_Q(v) = 1 - \mathbb{P}(Q > v)\end{aligned} \tag{2}$$

The fundamental theorem of calculus tells us

$$\int_{X \in \mathcal{D}} f_X(x)dx = F_X(x) + C \tag{3}$$

Using the survival-function/power-law-function, equation 1, and the tail function, equation 2, we can write

$$\frac{d}{dv}\left[F_Q(v) + C\right] = f_Q(v) \implies \frac{d}{dv}(1 - \mathbb{P}(Q > v) + C) = c\left(kv^{-(k+1)}\right) \tag{4}$$

For equation 4 to be a valid PDF is must be that

$$\int_{Q\in\mathcal{D}} f_Q(v)dv = 1 \tag{5}$$

3.1.1 Deriving Pareto Type 1 Distribution

We know the survival function is defined for all values greater than some minimum threshold that is $\{Q : v_m \leq v < \infty\}$.

$$\begin{aligned}
\int_{v_m}^{\infty} c\left(kv^{-k-1}\right) dv &= 1 \\
c\left[-\frac{k}{kv^k}\right]_{v_m}^{\infty} &= 1 \\
-c\left[\frac{1}{v^k}\right]_{v_m}^{\infty} &= 1 \\
-c\left[g(v=\infty) - g(v=v_m)\right] &= 1 \\
-c\left[0 - \frac{1}{v_m^k}\right] &= 1 \\
c &= v_m^k
\end{aligned} \tag{6}$$

Substituting in $c = v_m$ into the definition of the probability density function equation 4, we recover the **Pareto type 1** distribution with probability density 7.

$$f_Q(v) = \begin{cases} k\left(\frac{v_m^k}{v^{k+1}}\right) & , v \geq v_m \\ 0 & , v < v_m \end{cases} \tag{7}$$

Substituting $c = v_m$ into equation 2 yields the cumulative distribution function, written in equation 8.

$$F_Q(v) = \begin{cases} 1 - \left(\frac{v_m}{v}\right)^k & , v \geq v_m \\ 1 & , v < v_m \end{cases} \tag{8}$$

3.1.2 Optimal Parameters

We can derive the optimal parameter, k, using maximum likelihood estimation. The likelihood function is as follows.

$$\begin{aligned}
L(k;v) &= \prod_{i=1}^{n} f_Q(v) \\
&= \prod_{i=1}^{n} k\frac{v_m^k}{v_i^{k+1}} \\
&= \left(kv_m^k\right)^n \prod_{i=1}^{n} v_i^{-k-1} \\
\log(L(k;v)) &= n\log(k) + nk\log(v_m) - (k+1)\sum_{i=1}^{n}\log(v_i)
\end{aligned} \tag{9}$$

Taking derivatives with respect to the parameter k.

$$\frac{d}{dk}(\ell(k;v)) = \frac{n}{k} - n\log(v_m) - \sum_{i=1}^{n}\log(v_i) \tag{10}$$

Setting to zero and solving for $\hat{k}$, we have.

$$\frac{d}{dk}|_{k=\hat{k}} = 0 \implies \hat{k} = \frac{n}{\sum_{i=1}^{n}\log(v_i) + n\log(v_m)} \tag{11}$$

3.2 Trade Arrival

It is natural to think of the number of trades at any point in time as a counting process. Statistically, this is referred to as a **Poisson process**. Defining trade counts indexed by book side, $i \in \{a, b\}$, as $N_i(t)$, the set of $\{N_i(t=0), N_i(t=1), \ldots, N_i(t=T)\}$ is a Poisson process, where at each time we have the total number of trades at that time. Each increment of this process is a realisation of the Poisson distribution such that.

$$N_i(t+\Delta t) - N_i(t) \sim \text{Poisson}(\lambda \Delta t) \tag{12}$$

Importantly, each increment of a Poisson process is independent and identically distributed (*iid*).

If all increments of counts over time interval Δt are *iid*, then the frequency parameter λ, which tells us the expected number of counts per unit time Δt is constant. To estimate λ we again calculate the MLE.

$$L(\lambda; N_i(t)) = \prod_{j=1}^{n}\frac{(\lambda\Delta t)^{n_j}e^{\lambda\Delta t}}{n_j!} \tag{13}$$

Simplifying notation by letting $\Delta t = t$ and letting observations of the Poisson random variable be equal to k we have.

$$L(\lambda; N(t)) = \prod_{i=1}^{n}\frac{(\lambda t)^{k_i}e^{-\lambda t}}{k_i!} \tag{14}$$

$$\ell(\lambda; N(t)) = \sum_{i=1}^{n}\left[k_i\log(\lambda t) - \lambda t - \log(k_i!)\right] \tag{15}$$

$$= \log(\lambda t)\sum_{i=1}^{n}k_i - n\lambda t - n\log(k_i!) \tag{16}$$

Similar to equation 11, setting to 0 and solving for λ we have.

$$\frac{d}{d\lambda}(\ell(\lambda; N(t)))|_{\lambda=\hat{\lambda}} = 0 \implies \frac{1}{\lambda}\sum_{i=1}^{n}k_i - nt = 0 \implies \hat{\lambda} = \frac{\sum_{i=1}^{n}k_i}{nt} \tag{17}$$

Equation 17 tells us to compute the frequency parameter that is most probable under our data we calculate the sum of the number of counts per unit time t and divide it by the total time under our process. The frequency of trade arrivals per unit time is equal to the final Poisson process over T (total time).

3.2.1 The Poisson book

As we have hinted to earlier, by subscripting the Poisson-distributed random variable of trade counts $N(t)$, trades on each side of the book follow their own Poisson process. Formally, if events in a Poisson process each happen with some probability p_i, then $N_i(t)p_i$ is itself a Poisson process with frequency $\lambda_i t$. Therefore, we have two count processes in continuous time.

$$N_a(t) \sim \text{Poisson}(\hat{\lambda}_a t) \tag{18}$$

$$N_b(t) \sim \text{Poisson}(\hat{\lambda}_b t) \tag{19}$$

3.3 Volume and Queue Priority

We have learned we can extract new Poission processes from an original aggregate Poisson process if classes of events occur with some some probability p_i.

To motivate a Poisson process that combines **trade size** (chapter 3.1) and **trade arrival** (chapter 3.2), we first consider we have an agent who sets a limit order on both sides of the book; the agent knows in order for his order to be executed quickly, it must lie close to the fair value. The agent's queue position in the book is denoted as (m, k), which reads as the k'th order in the m'th level, where each level has n_i orders, for some side, s, in the book.

We will model fair value using the micro-price, p_m. Let p_a = price of the ask limit order and p_b = price of the bid limit order. In order for p_a to be filled $p_m \geq p_a$, as the limit ask guarantees a price at least as good as itself. Similarly, p_b is filled if and only if $p_m \leq p_b$. Determining the trade that will fill first can be used as predicative signal of asset price movement. For example, imagine the likelihood of waiting a maximum time t until a quote-clearing trade fills our limit order is more likely on the ask side. This means we can expect more aggressive selling at the bid that will shift the price down, hence we should go short.

$$V^{(s)}_{m,k} = \sum_{i=1}^{m-1} \sum_{j=1}^{n_i} V_{i,j} + \sum_{i=1}^{k} V^{(s)}_{m,i},\ 1 \leq k \leq m \tag{20}$$

Equation 20 tells us the resting volume with queue priority in the LOB for some limit order. Where s = the side of the book, m = the number of levels our order is away from the top of the book (TOB) and n_i = is the number of orders in each level. If $i = m$, this means we are looping over the level that contains our order. k denotes the position of our order in the price level. The first term simply calculates the total s side volume excluding the price level that contains our order, and the second term calculates the total volume with queue priority in the price level that contains our order.

3.4 Probabilistic Quote Clearing

Using equation 8 we can calculate the probability of observing a trade that arrives with enough volume to clear all levels and fill our order [equation 21].

$$\mathbb{P}(Q > V^{(s)}) = \left(\frac{v_m}{V^{(s)}}\right)^{\hat{k}} \tag{21}$$

By weighting frequencies of trade arrival on each side of the book using equations 18 and 19 we can derive the frequency of observing k quote-clearing trades per unit time t per side of the book.

$$\Lambda_a = \hat{\lambda}_a \mathbb{P}(Q > V^{(a)}) \tag{22}$$
$$\Lambda_b = \hat{\lambda}_b \mathbb{P}(Q > V^{(b)}) \tag{23}$$

So by Poisson thinning, we have a refined Poisson process with distribution parameters given by equations 22 and 23, which we will refer to as quote clearing intensities. So the number of trades on each side of the book that clear our quote per time increment Δt form a Poisson process given by equations 24 and 25.

$$NQ_a(t) \sim \text{Poisson}(\Lambda_a) \tag{24}$$
$$NQ_b(t) \sim \text{Poisson}(\Lambda_b) \tag{25}$$

3.5 Probabilistic Timing

We know that the time between observing the next Poisson event is exponentially distributed. This is equivalent to saying the waiting time, t, before observing the first Poisson event is exponentially distributed.

$$\text{waiting time until } (NQ^{(s)}(t) = 1) \sim \text{Exp}(\Lambda^{(s)}) \tag{26}$$

From the definition of the exponential distribution, it follows.

$$f(t; \Lambda^{(s)}) = \begin{cases} \Lambda^{(s)} e^{\Lambda^{(s)} t} & , t \geq 0, \\ 0 & , t < 0 \end{cases} \tag{27}$$

$$F(t; \Lambda^{(s)}) = \begin{cases} 1 - e^{-\Lambda^{(s)} t} & , t \geq 0 \\ 0 & , t < 0 \end{cases} \tag{28}$$

Where 27 denotes the probability density function and 28 denotes the cumulative distribution function.

4 Trading Premise

To trade the LOB we must recover snapshots. We assume we have $t = 1, 2 \ldots T$ snapshots. A property of the Poisson distribution is that events (observed counts in unit time) are *iid*, meaning each count $NQ_t^{(s)}$ is a random variable with its own associated Poisson distribution.

The reason for introducing the exponential distribution is so we can work within the time domain rather than the count domain (the signal is invariant). Because the Exponential CDF is an increasing function in $\Lambda^{(s)}$, the signal $\Lambda_a - \Lambda_b$ is identical to $F(t; \Lambda_a) - F(t; \Lambda_b)$.

$$\text{signal} = \text{sign}\left(F(t; \Lambda_a(t)) - F(t; \Lambda_b(t))\right) = \begin{cases} 1 & , \text{Long} \\ -1 & , \text{Short} \end{cases} \tag{29}$$

5 Data

As mentioned in 5.2 we use SPDR SPY L3 data, sourced from Databento [6]. Our Data is sourced from the XNAS venue –a venue, is defined as a exchange model, other examples include OTC markets and Dark Pools– from the XNAS:ITCH dataset, which disseminates full-depth-of book data for all its listings in the NASDAQ exchange [7]. Our specific dataset has 8304967 observations of 14 features spanning a time of 16 hours 54 minutes and 55 seconds, sampled from 2025-07-22 07:05:03 to 2025-07-22 23:59:59 (times in ISO 8601 format).

5.1 L3 Data

We specifically use a Market by Order (MBO) schema-specific data format–, which is a naming convention our data provider uses to define L3 market data. Level three (L3) data records all order events received by the trading venue. Each data record lists the price, order action and time of order arrival to the matching engine, among other metadata -data about our order data-, allowing us to determine queue position and build the LOB.

5.2 Modelling Fees

We must accurately model market frictions, most importantly, fees, so we can have a good idea of our models' live market performance. This is arguably the most important part of the back-test; therefore, much care was taken to model fees accurately. Importantly, we will be modeling fees from the perspective of a **HFT firm**, and therefore will not incur standard retail fees.

We derive our predictive features and test our model on the SPDR SPY Exchange Traded Fund (ETF). We were unable to source fee information for HFTs trading the SPY on NASDAQ, although given this security trades on more than one exchange we defaulted to its primary exchange, NYSE Arca. The SPDR SPY ETF trades for more than $1 and fits under the Exchange Traded Product (ETP) category for asset listed on NYSE Arca (an exchange for trading ETPs). Using NYSE Arca ETF trading fees [4], we implement a per share fee of $0.0030 [5].

5.2.1 Equity Curve

$$E_t = E_{t-1}\left(e^{r_t \times \text{signal}_t}\right) - \mathbb{1}\{c_t : \text{signal}_t \neq \text{signal}_{t-1}\},\ 1 \leq t \leq T \tag{30}$$

$$r_t = \ln\left(\frac{p_t}{p_{t-1}}\right),\ s_t = \frac{E_{t-1}}{p_t} \times 0.003 \times 2 \tag{31}$$

Equation 30 defines the equity curve of our model. Where $r_t =$ is the **instantaneous return** at time t and $s_t =$ is the market friction or fee incurred for trading $\frac{E_{t-1}}{p_t}$ shares at a cost of 0.003 per share [**??**]. The reason we multiply by two is that this strategy always holds a position in the market; to switch order position, the strategy must sell (buy) and then short (long) using all available account balance (as we also impose a full investment wealth constraint).

5.3 Building the Book (L3 to LOB)

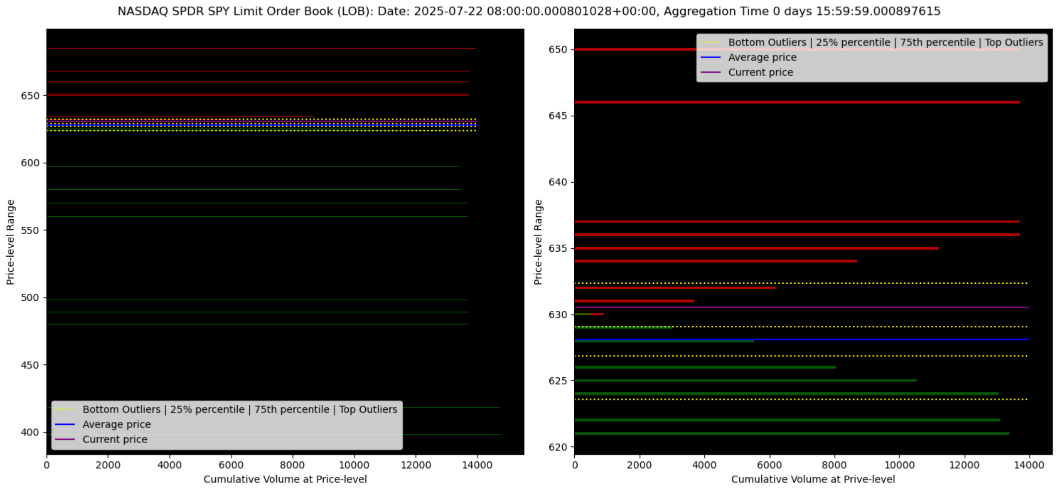

Figure 2: Global (left) and local (right) Limit order book (LOB) of SPDR SPY ETF trading on the NASDAQ exchange, using all orders received on the 22-07-2025.

Figure 2 is our dataset's LOB constructed using the FIFO matching-engine architecture outlined in 1.1.4. This is the visual equivalent of Table 1

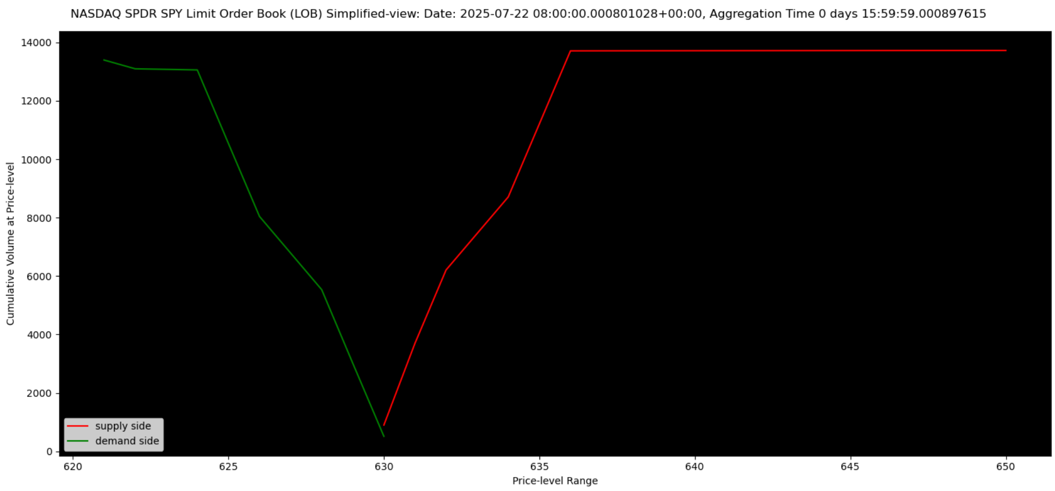

Figure 3: Local limit order book (LOB) of SPDR SPY ETF trading on the NASDAQ exchange in traditional graphical representation.

Figure 3 is a simplified representation of the full book in Figure 2. Both these figures convey identical information and have the same interpretation.

5.4 Volume Percentiles

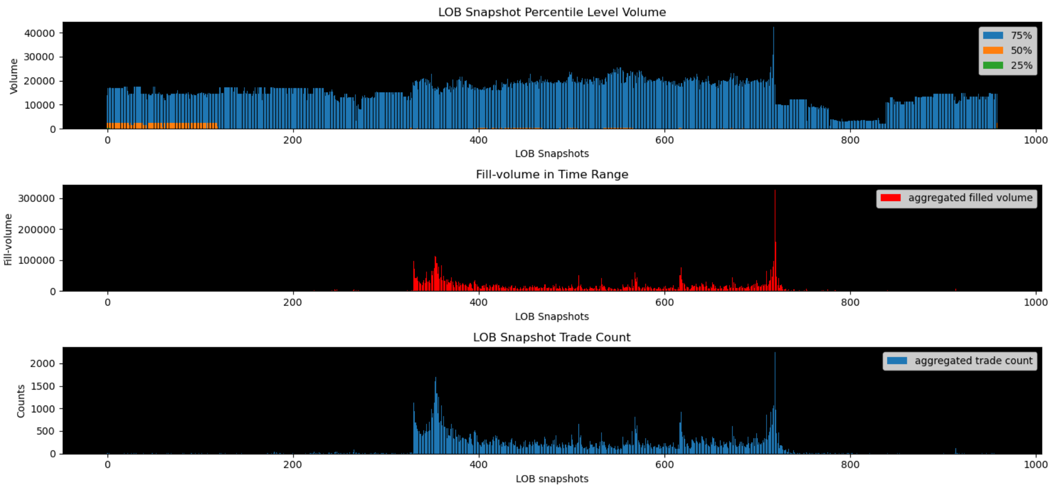

Figure 4: Percentile level volume (top), total volume liquidation (middle), and total trades (bottom) of NASDAQ SPDR SPY ETF limit order book (LOB) at 5-minute increments from "22-07-2025 08:00:00." to "22-07-2025 23:59:59.".

The 25%th, 50%th, and 75%th level volumes vary across per 5-minute LOB snapshot [Figure 3]. The blue bars dominate, suggesting volume concentration is not uniform across time. This means volume size across levels does not follow a linear relationship. This strengthens our hypothesis of using a heavy-tailed distribution to model trade size 3.1.

The evident spike of volume at "20:00:00" (book snapshot 718) signifies a large quantity of order arrivals. To determine if the large loss of resting order volume is the result of trade modifications or volume liquidations, we must look at filled volume per snapshot. In the case of heavy volume liquidation, causing the 75% percentile of volume to be wiped from the book, the order of resting volume and filled volume matters. If filled volume spikes after the 75% resting volume spike, this confirms that aggressive market participants cleared a large quantity of volume from the book.

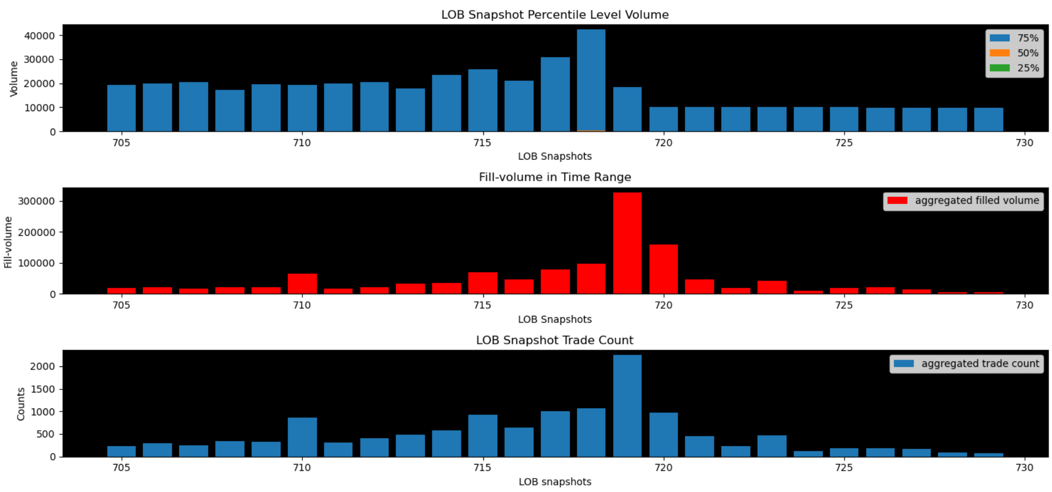

Figure 5: Percentile level volume (top), total volume liquidation (middle), and total trades (bottom) of NASDAQ SPDR SPY ETF limit order book (LOB) at 5-minute increments from "22-07-2025 19:46:00." to "22-07-2025 20:11:00".

Figure 4 is a zoomed-in version of figure 3. The number of aggressive market participants and fill volume coincide with a reduced 75% volume percentile [Figure 4]. This confirms that a high number of trades resulted in liquidity removal from the book; this is analogous to the example discussed in chapter 1.1.

6 Backtest Results

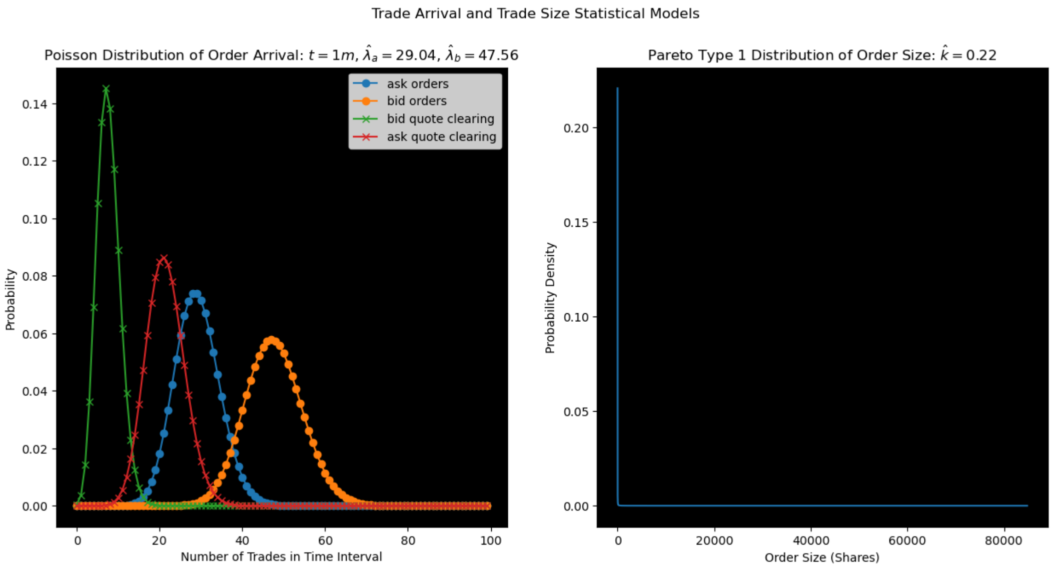

Figure 6: Parametric estimate of the trade arrival (left) and trade size (right) random variables.

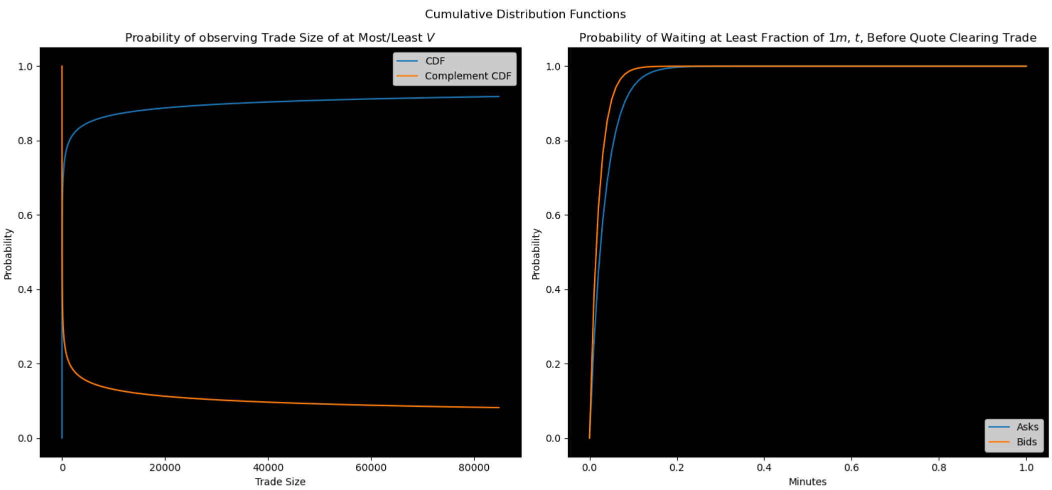

Figure 7: Cumulative distribution function and complement cumulative distribution function (survival function) of trade size (left) and cumulative distribution of time until first quote clearing trade arrival for bids and asks (right).

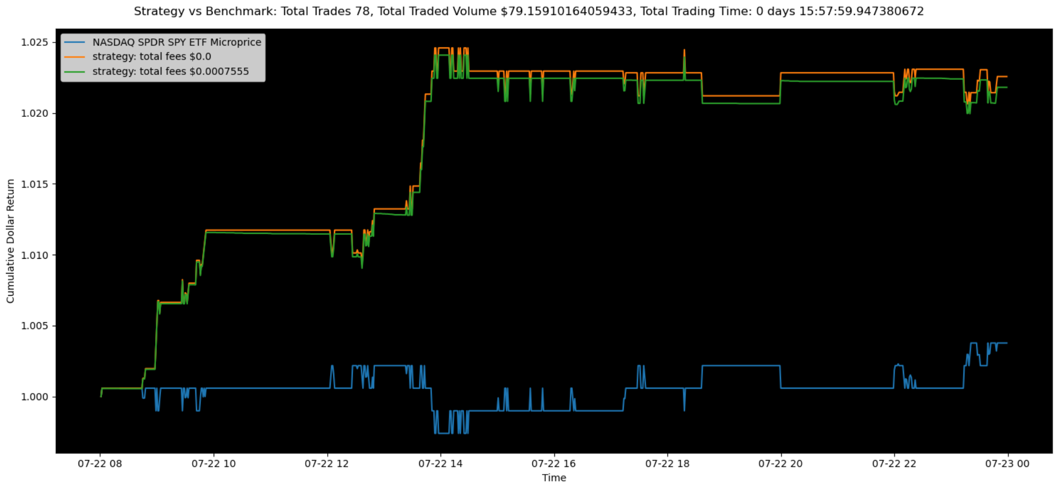

Figure 8: Benchmark vs strategy with and without trading fees.

7 Discussion

The left plot of Figure 6 compares the parametrised discrete Poisson probability density of the number of trades [chapter 3.2.1] and the number of quote clearing trades defined for each side of the book [chapter 3.4]. The probability mass functions (PMFs) parametrised by the maximum likelihood estimators (MLEs) given by equation 17 are denoted in blue for asks and orange for bids [Figure 7]. The ask MLE of a Poisson, $\lambda \in [0, \infty)$, tells us the expected number of events in unit time. The MLE for the number of ask trades in one minute is 29.04, and the MLE for bid trades in one minutes is 47.56, meaning given our data we can expect on average more trades to hit our bid in any one minute interval. More trades in the bid side mean their is more aggressive selling than buying. After weighting the Poisson events by probabilities of trades that clear our queue volume [chapter 3.4] we get a refined Poisson process given by the green and red PMF functions with cross markers. We weight the original MLE by the probability of observing a trade with enough volume to clear all trades up too and including our position in the queue. For simplicity we assume we hold orders with the highest queue priority, i.e. the first trade in the book. The expected number of quote clearing trades is greatest on the ask side [Figure 6]. This is because the required volume to clear our order on the bid side is 4000 shares, compared to only 4 shares on the bid side. So by the Pareto distribution survivial function [Figure 7] we weight the MLE expected number of trades by a much lower probability (for the bids side), explaining the large shift in the PMF (orange shifts more than blue).

The right plot is a parametrised continuous probability density function of incoming trade side [equation 7] evaluated at different trade sizes [Figure 6]. The survival function of a Pareto distribution follows a power law, meanf for some parameter values the expected value is non finite, specifically for $k < 1$, given $\hat{k} = 0.22$, the expected trade size is infinite. In other words as the sample size of the number of trades grows the average trade size does not converge to any finite value.

Both plots in Figure 7 compare the cumulative probability density of their respective distributions. The left plot graphs the area to the left-hand side of the Pareto density for trade size given by equation 7 (chapter 3).This is interpreted as the likelihood of observing a value at most as extreme as the given trade size. The complement CDF (orange) is called the survival function and tells the probability of getting a trade size at least as large as our observation [Figure 7].

The plot to the right in Figure 8, graphs the likelihood of waiting a fraction of one minute until observing a trade on both sides of the limit order book. These are the exponential CDF functions we derive from the Poisson and they tell us the probability of waiting at most time t for a new trade on either side of the book. Figure 7 suggests that we expect to wait less time for a trade on the ask side than the bid side, as the CDF for the bid (orange) is greater than the CDF for asks (blue). This aligns with our findings from Figure 6, a larger expected number of trades (higher frequency parameter) means that we should expect shorted average waiting times between events. The mean waiting time is given by the expected value of an exponential, $\frac{1}{\lambda}$, which is decreasing in λ which confirms our intuition, that higher λ implies lower wait time between events.

We run our trading strategy on the entire dataset containing 15 hours LOB data, generating signals at the end of each one-minute batch. Figure 8 plots the cumulative returns for the benchmarks and trading strategy, with and without trading fees. Our strategy outperforms the benchmark from times $8:44$ to the end of the trading session at $23:59$, yielding a final cumulative return of 2.26%, over-performing the benchmark by 1.88% [].

8 Research Limitations

Drawbacks of our research include, although they are not limited to ...

Relying on Trade Size as a Predictor of Market Direction. We form our trading signal by weighting the Poisson process of trade arrival by the probability of receiving a quote-clearing trade in some time interval (one minute), going long if the frequency of ask clearing trades is greater than the frequency of bid clearing trades, and short otherwise. The logic is that a higher expected number of trades that hit ask means there must be more aggressive buyers than sellers, with the reverse being true of the bid case. The problem is that we are assuming trade-arrival and trade size follow some distribution, and basing signals using these assumptions, neglecting real market dominance indicators such as **LOB Imbalance** or **LOB Volume Weighted Imbalance**, which tell us the difference between bid and ask volume in the market.

9 Conclusion

Chapter 1 focused on explaining how buyers and sellers are matched in the limit order book (LOB). We defined various concepts, such as price taking, price making, limit orders, liquidity, and the bid-ask spread, and learned how market makers facilitate buy and sell orders at the top of the book (TOB).

We defined our research problem in chapter 2, which was to derive a predictive signal from the LOB. We took a probabilistic modelling approach to the task by modelling the processes of trade arrival and trade size. We derived a Pareto Type 1 distribution for trade size by first making an assumption that the survival function of this distribution follows a power law. Trade arrival was modelled using a Poisson distribution, which we broke up into bid and ask Poisson distributions using the independence of events property. We treated quote-clearing trades (trades with enough volume to fill our trade in the queue) as another independent event to further break up our Poisson distribution, using the trade size distribution to calculate the class probabilities. The result was a trade size weighted trade arrival distribution [chapter 3.4]. To define our trading signal we first transformed the intensities (Λ_a, Λ_b) into the probability domain using the exponential distribution. Our signal was long if the probability of waiting at most one minute for a quote clearing ask trade was greatest. This means all things held constant, one minute from now, a quote quote clearing trade that hits the ask side of the book is more likely, meaning ask trade intensity is higher, signalling higher demand than supply in the book.

Chapter 5 focused on data and fees. We used NASDAQ Level Three data to build our LOB. We also assumed we play the role of a large institution and therefore explored methods for simulating real-life trading for more accurate backtesting results. We conducted EDA on our constructed LOB using the FIFO databento matching engine in chapters 5.3 and 5.4 confirming a non-linear relationship in trade size counts, suggesting that making a power law distribution on trade size likelihood was rational.

We discussed our backtest results (chapter 6) in chapter 7 and found evidence that our trading strategy has an edge as it outperformed the benchmark by 1.88% in the training set. Despite strong in-sample performance, a decision on the success of this strategy cannot be made without multi-asset backtests. Our probabilistic approach to modelling the LOB has laid a strong foundation for more advanced models and the use of demand vs supply (imbalance) metrics, such as book imbalance (chapter 8) leaves the current model with a lot of potential for improvement.

10 Appendix

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